

US Patent & Trademark Office

Patent Public Search | Text View

United States Reissue Patent
Kind Code
Date of Reissued Patent
Inventor(s)

RE50570
E
September 02, 2025
Hong; Dae Jin et al.

Refrigerator

Abstract

An object of the present disclosure is to provide a refrigerator configured to improve main body stiffness, which is decreased as thickness of an insulation material is reduced to increase inner capacity of the main body, using a reinforcement structure, resulting in reduction of deformation of the main body. Another object of the present disclosure is to provide a refrigerator in which an electronic box including various electronic constituent elements needed to control the operation of the refrigerator is disposed in a hinge cover provided in a forward direction of an upper part of the main body, resulting in improved space utilization (or space occupancy). Another object of the present disclosure is to provide a refrigerator in which a reinforcement plate formed of steel is contained in the electronic box, such that, although a fire breaks out in constituent elements contained in the electronic box, the refrigerator can prevent the fire from spreading to the outside of the electronic box using the steel reinforcement plate.

Inventors: Hong; Dae Jin (Suwon-si, KR), Kim; Yong Woo (Suwon-si, KR)
Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)
Family ID: 53524110
Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)
Appl. No.: 17/245507
Filed (or PCT Filed): January 07, 2015
PCT No.: PCT/KR2015/000157
PCT Pub. No.: WO2015/105333
PCT Pub. Date: July 16, 2015

Foreign Application Priority Data

KR 10-2014-0002010

Jan. 07, 2014

Related U.S. Application Data

reissue parent-doc US 15110213 20150107 GRANTED US 10345036 20190709 child-doc US 17245507

Publication Classification

Int. Cl.: F25D29/00 (20060101); F25D11/02 (20060101); F25D17/06 (20060101); F25D23/02 (20060101); F25D23/06 (20060101); F25D25/02 (20060101)

U.S. Cl.:

CPC F25D29/005 (20130101); F25D11/022 (20130101); F25D17/065 (20130101); F25D23/028 (20130101); F25D23/062 (20130101); F25D25/025 (20130101); F25D2201/126 (20130101); F25D2323/021 (20130101); F25D2323/024 (20130101)

Field of Classification Search

CPC: F25D (17/065); F25D (17/04); F25D (17/045); F25D (17/06); F25D (17/062); F25D (23/062); F25D (2323/021); F25D (2323/024); F25D (11/022); F25D (23/028); F25D (25/025); F25D (29/005); F25D (2201/126)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2944410	12/1959	Mann	62/289	F25D 23/04
3004401	12/1960	Mann	62/156	F25D 17/065
3027732	12/1961	Mann et al.	N/A	N/A
3793847	12/1973	Scarlett	62/190	F25D 17/065
3893307	12/1974	Jacobs	62/150	F25D 21/14
5799500	12/1997	Kang	62/407	F25D 17/045
6347530	12/2001	Kim	N/A	N/A
7107785	12/2005	Bärmann et al.	N/A	N/A
8926034	12/2014	Park	N/A	N/A
2005/0126206	12/2004	Oh	N/A	N/A
2006/0070395	12/2005	Lee	62/408	F25D 17/045
2006/0096312	12/2005	Shim	62/419	F25D 17/065
2006/0248915	12/2005	Lee	62/407	F25D 11/022
2007/0044498	12/2006	Kang	62/276	F25D 11/022
2010/0125365	12/2009	Ahn	700/275	F25D 29/00
2011/0011118	12/2010	Cho	62/515	F25D 17/067
2011/0252817	12/2010	Tunzi	62/89	F25D 17/065
2012/0096887	12/2011	Cho	62/276	F25D 21/08
2013/0106274	12/2012	Yang	N/A	N/A
2020/0386468	12/2019	Matsuno	N/A	F25D 17/045

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1124056	12/1995	CN	N/A
1156238	12/1996	CN	N/A
1169184	12/1996	CN	N/A
1769823	12/2005	CN	N/A
2828688	12/2005	CN	N/A
1920452	12/2006	CN	N/A
102032744	12/2010	CN	N/A
202485306	12/2011	CN	N/A
202573314	12/2011	CN	N/A
202835993	12/2012	CN	N/A
103033015	12/2012	CN	N/A
202853233	12/2012	CN	N/A
203454520	12/2013	CN	N/A
11 2005 002 832	12/2007	DE	N/A
1904797	12/2011	EP	N/A
3 093 589	12/2015	EP	N/A
3 657 106	12/2019	EP	N/A
3 657 106	12/2021	EP	N/A
2 012 937	12/1978	GB	N/A
4-98072	12/1991	JP	N/A
8-152254	12/1995	JP	N/A
8-296953	12/1995	JP	N/A
10-103844	12/1997	JP	N/A
2000-346531	12/1999	JP	N/A
2002-90028	12/2001	JP	N/A
2002090028	12/2001	JP	N/A
2003-21448	12/2002	JP	N/A
2005-90924	12/2004	JP	N/A
2007-225193	12/2006	JP	N/A
2008-75890	12/2007	JP	N/A
2011-64340	12/2010	JP	N/A
2013-195030	12/2012	JP	N/A
10-2007-0043446	12/2006	KR	N/A
10-0800195	12/2007	KR	N/A
10-2010-0004810	12/2009	KR	N/A
10-2013-0057618	12/2012	KR	N/A
20120002042	12/2012	KR	N/A
101 45 141	12/2002	WO	N/A
WO 2006/120058	12/2005	WO	N/A
WO 2006/120076	12/2005	WO	N/A
WO 2007/006113	12/2006	WO	N/A
WO 2010/008141	12/2009	WO	N/A
WO 2010/008141	12/2009	WO	N/A

OTHER PUBLICATIONS

European Office Action dated May 16, 2023 in European Patent Application No. 20150463.6. cited by applicant

Extended European Search Report dated May 22, 2023 in European Patent Application No. 23160123.8. cited by applicant
“Versilbern” Wikipedia; Anlage 11 D5; Jan. 18, 2023; pp. 1 of 3. cited by applicant
European Office Action dated May 23, 2022 in European Patent Application No. 21 188 526.4. cited by applicant
Chinese Office Action for Chinese Patent Application No. 202011388084.2 dated Dec. 2, 2021. cited by applicant
Chinese Office Action for Chinese Patent Application No. 202011384101.5 dated Nov. 30, 2021. cited by applicant
European Search Report for European Patent Application No. 21188526.4 dated Oct. 19, 2021. cited by applicant
Office Action dated Dec. 12, 2022 in European Patent Application No. 21 188 526.4 (90 pages). cited by applicant
International Search Report mailed Apr. 13, 2015 in corresponding International Application No. PCT/KR2015/000157. cited by applicant
Written Opinion mailed Apr. 13, 2015 in corresponding International Application No. PCT/KR2015/000157. cited by applicant
Office Action issued Feb. 26, 2018, in corresponding Chinese Patent Application No. 201580010475.1, 17 pgs. cited by applicant
Chinese Office Action dated April 9, 2019 in Chinese Patent Application No. 201580010475.1. cited by applicant
Chinese Office Action dated September 1, 2020 in Chinese Patent Application No. 201580010475.1. cited by applicant
US Office Action dated Oct. 13, 2017 in U.S. Appl. No. 15/110,213. cited by applicant
US Office Action dated Mar. 1, 2018 U.S. Appl. No. 15/110,213. cited by applicant
US Office Action dated Jun. 7, 2018 in U.S. Appl. No. 15/110,213. cited by applicant
US Office Action dated Dec. 13, 2018 in U.S. Appl. No. 15/110,213. cited by applicant
US Notice of Allowance dated Mar. 1, 2019 in U.S. Appl. No. 15/110,213. cited by applicant
Brief Communication dated Mar. 12, 2024 including letter from the opponent 01 dated Mar. 4, 2024 in corresponding European application No. 20150463.6. cited by applicant
Brief Communication dated May 13, 2024 including EPO Form 2310, summons to attend oral proceedings pursuant to Rule 115(1) EPC in corresponding European application No. 20150463.6. cited by applicant
European Patent Office Brief Communication for Application 20150463.6 dated Feb. 14, 2025 regarding Letter from the opponent of Feb. 11, 2025. cited by applicant
European Office Action issued Feb. 5, 2025 for European Application No. 23 160 123.8. cited by applicant
Extended European Search Report dated Aug. 11, 2017 in corresponding European Patent Application No. 15735174.3. cited by applicant
Chinese Office Action dated Sep. 27, 2018 in Chinese Patent Application No. 201580010475.1. cited by applicant
Korean Office Action dated Jan. 15, 2019 in Korean Patent Application No. 10-2014-0089516. cited by applicant

Primary Examiner: Gilbert; William V

Attorney, Agent or Firm: STAAS & HALSEY LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a .Iadd.broadening reissue application of U.S. Pat. No. 10,345,036, issued Jul. 9, 2019, which is a .Iaddend.U.S. National Stage Application, which claims the benefit under 35 U.S.C. § 371 of PCT International Patent Application No. PCT/KR2015/000157, filed Jan. 7, 2015 which claims the foreign priority benefit under 35 U.S.C. § 119 of Korean Patent Application No. 10-2014-0002010, filed Jan. 7, 2014, and Korean Patent Application No. 10-2014-0089516, filed Jul. 16, 2014, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) Embodiments of the present disclosure relate to a refrigerator having an improved cool air duct so as to uniformly maintain a temperature of an indoor space of a storage chamber.

BACKGROUND ART

(3) Generally, a refrigerator includes a main body having an inner casing and an outer casing, a storage chamber formed by the inner space, and a cool air supply device to supply cool air to the storage chamber, to thereby store foods in a fresh state.

(4) A temperature of the storage chamber is maintained within a predetermined range needed to store foods in a fresh state.

(5) The storage chamber of the refrigerator has an opened front surface, and the opened front surface is closed by a door in such a manner that a temperature of the storage chamber is properly maintained at ordinary times.

(6) The insulation is foamed between the inner casing and the outer casing of the main body so as to prevent leakage of cool air from the storage chamber.

(7) Since the insulation is foamed at a predetermined temperature or higher, heat occurs in the insulation foaming process, a temperature of the main body may be higher than a room temperature (or normal temperature) by about 20° C. or higher on the condition that the insulation is foamed between the inner casing and the outer casing.

(8) After the insulation is foamed between the inner casing and the outer casing, as a temperature of the main body gradually drops to a room (or normal) temperature, the insulation is hardened or stiffened, and the main body is thermally contracted.

(9) The inner casing may be generally formed of a plastic material, and the outer casing may be generally formed of steel. The degree of thermal contraction of the plastic material may be higher than that of the steel material by about 5 times or higher. As a result, when the main body is thermally contracted, the inner casing is thermally contracted much more than the outer casing. Thus, during a predetermined time in which the temperature of the main body drops to room temperature, the center part of both sides of the main body is modified into a convex shape in an outward direction of the main body. Under the condition that the temperature of the main body drops to the room temperature, the center part of both sides of the main body is modified into a convex shape in an outward direction of the main body such that the insulation is hardened or stiffened.

(10) When deformation occurs in the inner casing and the outer casing due to a difference in thermal contraction between the inner casing and the outer casing, such deformation of the inner casing and the outer casing may be regularly reduced by the insulation contacting each of the inner casing and the outer casing. If thickness of the insulation foamed between the inner casing and the outer casing is reduced so as to implement the same exterior size as well as to increase inner capacity (or inner volume) of the main body, the degree of deformation in which the center part of both sides of the main body is modified into a convex shape in an outward direction of the main body may unavoidably increase. When a refrigerator starts operation upon completion of insulation

foam, the internal temperature of the main body is lowered and at the same time the degree of thermal contraction of the inner casing is increased, resulting in an increase in the degree of deformation of the exterior shape.

(11) In addition, if thickness of the insulation is reduced, insulation performance may be deteriorated. Stiffness or rigidity of the inner casing and the outer casing is weakened, such that the main body may be modified in shape not only by a weight of the main body but also by a weight of materials stored in the main body.

(12) In order to improve deteriorated insulation performance caused by a reduced thickness of the insulation, not only the insulation but also a vacuum insulation may be disposed between the inner casing and the outer casing. However, although the vacuum insulation and the insulation are disposed between the inner casing and the outer casing, it is impossible to reinforce weakened stiffness whereas insulation performance is improved.

DISCLOSURE

Technical Problem

(13) An object of the present disclosure is to provide a refrigerator configured to improve main body stiffness, which is decreased as thickness of an insulation material is reduced to increase inner capacity of the main body, using a reinforcement structure, resulting in reduction of deformation of the main body.

(14) Another object of the present disclosure is to provide a refrigerator in which an electronic box including various electronic constituent elements needed to control the operation of the refrigerator is disposed in a hinge cover provided in a forward direction of an upper part of the main body, resulting in improved space utilization (or space occupancy).

(15) Another object of the present disclosure is to provide a refrigerator in which a reinforcement plate formed of steel is contained in the electronic box, such that, although a fire breaks out in constituent elements contained in the electronic box, the refrigerator can prevent the fire from spreading to the outside of the electronic box using the steel reinforcement plate.

(16) Another object of the present disclosure is to provide a refrigerator including an improved cool air duct so as to uniformly maintain a temperature of an indoor space of a storage chamber.

Technical Solution

(17) Accordingly, the present disclosure is directed to a refrigerator that substantially obviates one or more problems due to limitations and disadvantages of the related art.

(18) In accordance with one aspect of the present disclosure, a refrigerator includes a main body, a storage chamber provided in the main body in a manner that a front surface of the storage chamber is opened, an evaporator to supply cool air to the storage chamber, and a cool air duct to form a flow passage in a manner that cool air generated by the evaporator is supplied to the storage chamber, wherein the cool air duct includes a front panel formed of a metal material, which is provided at a back surface of the storage chamber, such that an indoor space of the storage chamber is evenly cooled by cool air of the indoor space of the storage chamber and a uniform temperature is thus maintained in the indoor space of the storage chamber, wherein an upper part of the front panel is formed in a flat panel shape, and a lower part of the front panel is formed in a streamlined shape that is gradually curved in a direction toward a front surface of an upper part of the storage chamber as the lower part of the front panel approaches a lower end of the front panel.

(19) The cool air duct may include the front panel, a cool air flow passage unit disposed at a back surface of the front panel so as to form a first flow passage, a blowing fan mounting unit provided at a lower part of the cool air flow passage unit so as to supply cool air generated by the evaporator to the storage chamber.

(20) The front panel may be configured to have a width corresponding to a width of the storage chamber.

(21) The front panel may be formed of aluminum (Al), and include a plurality of cool air outlets through which the cool air guided through the flow passage is discharged to the inside of the

storage chamber.

(22) The blowing fan may be arranged at an upper part of the evaporator, the lower part of the front panel is formed in a streamlined shape so as to accommodate an installation space of the blowing fan.

(23) The lower part of the front panel may include a barrier to form a space in which the evaporator and the blowing fan are installed, and the barrier closely adheres to the lower part of the front panel such that the storage chamber and the space in which the evaporator and the blowing fan are installed are sealed up.

(24) The cool air flow passage unit may include a first cool air flow passage unit disposed at a back surface of the front panel, and a second cool air flow passage unit connected to a back surface of the first cool air flow passage unit such that the flow passage is formed between the first cool air flow passage unit and the second cool air flow passage unit.

(25) The first cool air flow passage unit may include a plurality of discharge holes located at positions corresponding to positions of the plurality of cool air outlets.

(26) The blowing fan mounting unit may be located at a lower part of the cool air flow passage unit, and include a housing to which the blowing fan is rotatably mounted and a cover member to cover an opened front surface of the housing.

(27) A drain unit may configure to drain condensed water generated from the evaporator to the outside is disposed at a lower part of the evaporator.

(28) The drain unit may have a tilted surface that is gradually tilted downward with decreasing distance from a right end of the drain unit on the basis of a center part of the drain unit, and a drain hole through which the condensed water is drained is provided at an end of the tilted surface.

(29) The drain hole may include a drain pipe through which the condensed water is drained to the outside of the main body, and the drain pipe is disposed between an inner casing and an outer casing of a side surface of the main body.

(30) In accordance with another aspect of the present disclosure, a refrigerator includes a main body, a storage chamber provided in the main body in a manner that a front surface of the storage chamber is opened, an evaporator to supply cool air to the storage chamber, a blowing fan to guide the cool air generated by the evaporator such that the cool air is supplied to the storage chamber, and a cool air duct to form a flow passage in a manner that the cool air generated by the evaporator is supplied to the storage chamber, wherein the cool air duct includes a front panel formed of a metal material, which is provided at a back surface of the storage chamber, such that an indoor space of the storage chamber is evenly cooled by cool air of the indoor space of the storage chamber and a uniform temperature is thus maintained in the indoor space of the storage chamber, a cool air flow passage unit disposed at a back surface of the front panel so as to form the flow passage, and a blowing fan mounting unit that is provided at a lower part of the cool air flow passage unit, is connected to the blowing fan, and is formed to protrude farther forward than the cool air flow passage unit.

(31) The front panel may be formed of aluminum (Al), and may have a width corresponding to a width of the storage chamber.

(32) The cool air flow passage unit may include a first cool air flow passage unit disposed at a back surface of the front panel, and a second cool air flow passage unit connected to a back surface of the first cool air flow passage unit such that the flow passage is formed between the first cool air flow passage unit and the second cool air flow passage unit.

(33) A drain unit may configure to drain condensed water generated from the evaporator to the outside is disposed at a lower part of the evaporator.

(34) The drain unit may have a tilted surface that is gradually tilted downward with decreasing distance from a right end of the drain unit on the basis of a center part of the drain unit, and a drain hole through which the condensed water is drained is provided at an end of the tilted surface.

(35) The drain hole may include a drain pipe through which the condensed water is drained to the

outside of the main body,

(36) wherein the drain pipe is disposed between an inner casing and an outer casing of a side surface of the main body.

(37) In accordance with another aspect of the present disclosure, A refrigerator includes a main body, a storage chamber provided in the main body in a manner that a front surface of the storage chamber is opened, an evaporator to supply cool air to the storage chamber, a blowing fan to guide the cool air generated by the evaporator such that the cool air is supplied to the storage chamber, and a cool air duct to form a flow passage in a manner that the cool air generated by the evaporator is supplied to the storage chamber, wherein the cool air duct includes a front panel formed of a metal material having a width corresponding to a width of the storage chamber, which is provided at a back surface of the storage chamber such that an indoor space of the storage chamber is evenly cooled by cool air of the indoor space of the storage chamber and a uniform temperature is thus maintained in the indoor space of the storage chamber, a cool air flow passage unit disposed at a back surface of the front panel so as to form the flow passage, and a blowing fan mounting unit that is provided at a lower part of the cool air flow passage unit, is connected to the blowing fan, and is formed to protrude farther forward than the cool air flow passage unit.

Advantageous Effects

(38) As is apparent from the above description, the refrigerator according to the embodiments can maintain stiffness or rigidity using a reinforcement structure even when thickness of an insulation material is reduced, resulting in reduction of deformation of a main body of the refrigerator.

(39) An electronic box is disposed in a hinge cover, such that space utilization is improved and a fire occurring in an electronic box is prevented from spreading to the outside of the electronic box.

(40) A heat dissipation pipe is located at the position adjacent to the outer casing of the refrigerator, such that the refrigerator can prevent dew formation and the heat dissipation pipe can be easily fixed to the inner casing of the refrigerator.

(41) A temperature of the indoor space of the storage chamber can be uniformly maintained, resulting in energy savings.

Description

DESCRIPTION OF DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the invention, illustrate embodiments of the invention and together with the description serve to explain the principle of the invention.

(2) FIG. 1 is a perspective view illustrating a refrigerator according to an embodiment of the present disclosure.

(3) FIG. 2 is a cross-sectional view illustrating a side of the refrigerator according to an embodiment of the present disclosure.

(4) FIG. 3 is a front view illustrating the refrigerator according to an embodiment of the present disclosure.

(5) FIG. 4 is a view illustrating a refrigerator in which a reinforcement member is attached to an inner casing according to an embodiment of the present disclosure.

(6) FIG. 5 is a cross-sectional view illustrating a refrigerator in which a first reinforcement member is attached to the inner casing according to an embodiment of the present disclosure.

(7) FIG. 6 is a view illustrating a refrigerator in which a reinforcement member is attached to the outer casing according to an embodiment of the present disclosure.

(8) FIG. 7 is a view illustrating a refrigerator in which a reinforcement member is vertically attached to the inner casing according to an embodiment of the present disclosure.

(9) FIG. 8 is a view illustrating a refrigerator in which a reinforcement frame is coupled to a main

body according to an embodiment of the present disclosure.

(10) FIG. **9** is a perspective view illustrating a reinforcement frame according to an embodiment of the present disclosure.

(11) FIG. **10** is an exploded perspective view illustrating an electronic box embedded in a refrigerator according to an embodiment of the present disclosure.

(12) FIG. **11** is an exploded perspective view illustrating the electronic box viewed from the standpoint of a lower part of the refrigerator according to an embodiment of the present disclosure.

(13) FIG. **12** is a perspective view illustrating an electronic box according to an embodiment of the present disclosure.

(14) FIG. **13** is a cross-sectional view illustrating an electronic box mounted to the main body according to an embodiment of the present disclosure.

(15) FIG. **14** is a view illustrating a wire connected to the electronic box according to an embodiment of the present disclosure.

(16) FIG. **15** is a view illustrating that a heat dissipation pipe is disposed in the main body according to an embodiment of the present disclosure.

(17) FIG. **16** is a view illustrating an inner casing including a heat dissipation pipe and an outer casing according to an embodiment of the present disclosure.

(18) FIG. **17** is a view illustrating that the heat dissipation pipe is fixed to the inner casing according to an embodiment of the present disclosure.

(19) FIG. **18** is a view illustrating that a seating unit for seating the heat dissipation pipe therein and fixing grooves for fixing the heat dissipation pipe are mounted to the inner casing according to an embodiment of the present disclosure.

(20) FIG. **19** is a view illustrating that the heat dissipation pipe is disposed in the main body according to an embodiment of the present disclosure.

(21) FIG. **20** is a view illustrating a storage unit disposed in a storage chamber according to an embodiment of the present disclosure.

(22) FIG. **21** is a view illustrating that a sliding shelf is coupled to the inside of the storage chamber according to an embodiment of the present disclosure.

(23) FIG. **22** is a view illustrating that a sliding shelf is coupled to the inside of the storage chamber according to an embodiment of the present disclosure.

(24) FIG. **23** is a view illustrating that a first storage box is coupled to the sliding shelf according to an embodiment of the present disclosure.

(25) FIG. **24** is an enlarged view illustrating that a rail cover of FIG. **23** is connected to a coupling unit.

(26) FIG. **25** is a view illustrating that the sliding shelf is coupled to the first storage box according to an embodiment of the present disclosure.

(27) FIG. **26** is a view illustrating that the sliding shelf viewed from the standpoint of a bottom surface of the refrigerator according to an embodiment of the present disclosure.

(28) FIG. **27** is a view illustrating that a slide unit is pulled out from the sliding shelf shown in FIG. **26**.

(29) FIG. **28** is an exploded perspective view illustrating a self-closing device according to an embodiment of the present disclosure.

(30) FIG. **29** is a view illustrating the self-closing device according to an embodiment of the present disclosure.

(31) FIG. **30** is a view illustrating some parts of the self-closing device viewed from the bottom surface of the refrigerator according to an embodiment of the present disclosure.

(32) FIG. **31** is a view illustrating that a first storage box and a second storage box are detached from each other according to an embodiment of the present disclosure.

(33) FIG. **32** is a side view illustrating a storage unit according to an embodiment of the present disclosure.

- (34) FIG. 33 is a view illustrating that the second storage box of FIG. 32 moves from one place to another place.
- (35) FIG. 34 is a view illustrating that the second storage box is disposed in the first storage box according to an embodiment of the present disclosure.
- (36) FIG. 35 is a view illustrating a shelf unit according to an embodiment of the present disclosure.
- (37) FIG. 36 is a view illustrating that a first shelf of FIG. 35 is detached from a support unit.
- (38) FIG. 37 is a view illustrating that a horizontal maintenance unit is connected to a bracket according to an embodiment of the present disclosure.
- (39) FIG. 38 is a view illustrating that a horizontal maintenance unit is connected to a shelf according to an embodiment of the present disclosure.
- (40) FIG. 39 is a view illustrating that a fixing protrusion is inserted into a fixing groove according to an embodiment of the present disclosure.
- (41) FIG. 40 is a view illustrating the inside of an upper storage chamber according to an embodiment of the present disclosure.
- (42) FIG. 41 is an exploded perspective view illustrating a first cool air duct according to an embodiment of the present disclosure.
- (43) FIG. 42 is a view illustrating that the first cool air duct is disposed in the refrigerator according to an embodiment of the present disclosure.
- (44) FIG. 43 is a view illustrating that a linear guide member is disposed in the refrigerator according to an embodiment of the present disclosure.
- (45) FIG. 44 is a view illustrating that the linear guide member of FIG. 43 is connected to an insulation inlet mounted to a cover of a machine room.
- (46) FIG. 45 is a view illustrating that another exemplary guide member of FIG. 44 is connected to an insulation inlet mounted to a cover of a machine room.
- (47) FIG. 46 is a view illustrating that a Y-shaped guide member is disposed in the refrigerator according to an embodiment of the present disclosure.
- (48) FIG. 47 is a view illustrating that the Y-shaped guide member of FIG. 46 is connected to the insulation inlet mounted to the machine room cover.
- (49) FIG. 48 is a view illustrating that another exemplary guide member of FIG. 47 is connected to the insulation inlet mounted to the machine room cover.
- (50) FIG. 49 is a view illustrating that a refrigerant pipe and a drain pipe are arranged at a side of the main body of the refrigerator according to an embodiment of the present disclosure.

BEST MODE

- (51) Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout.
- (52) Referring to FIGS. 1 to 3, the refrigerator may include a main body 10; a plurality of storage chambers 20 each having an opened front surface in the main body 10; a door 30 rotatably coupled to the main body 10 so as to open or close the storage chambers 20; and a hinge unit 40 through which the door 30 is rotatably coupled to the main body 10.
- (53) The main body 10 may include an inner casing 11 to form the storage chambers 20; an outer casing 13 to form the exterior appearance of the refrigerator; and a cool air supply device (not shown) to provide the cool air to the storage chambers 20.
- (54) The cool air supply device may include a compressor C, a condenser (not shown), an expansion valve (not shown), an evaporator E, a blowing fan F, a cool air duct D, etc. The insulation 15 is foamed between the inner casing 11 and the outer casing 13 of the main body 10 so as to prevent leakage of cool air from the storage chambers 20.
- (55) The compressor C, the condenser, the expansion valve, and the evaporator E may be connected through a refrigerant pipe P, and refrigerant may be directed through the refrigerant pipe P.

(56) A machine room **28** may be provided at a rear lower portion of the main body **10**, in which the compressor **C** to compress refrigerant and the condenser to condense the compressed refrigerant are mounted.

(57) The evaporator **E** may include a first evaporator **E1** to supply cool air to an upper storage chamber **21** and a second evaporator **E2** to supply cool air to a lower storage chamber **23**. The cool air generated by the first evaporator **E1** may be supplied to the upper storage chamber **21** through a first blowing fan **F1**, and the cool air generated by the second evaporator **E2** may be supplied to the lower storage chamber **23** through a second blowing fan **F2**.

(58) The cool air duct **D** may include a first cool air duct **700** mounted to a back surface of the upper storage chamber **21**, and a second cool air duct **760** mounted to a back surface of the lower storage chamber **23**. The first cool air duct **700** may form a first flow passage through which cool air generated by the first evaporator **E1** is supplied to the upper storage chamber **21** through the first blowing fan **F1**. The second cool air duct **760** may form a second flow passage **763** through which cool air generated by the second evaporator **E2** is supplied to the lower storage chamber **23** through the second blowing fan **F2**.

(59) A first cool air outlet **711** may be disposed in the first cool air duct **700**, such that cool air generated by the first evaporator **E1** is supplied to the upper storage chamber **21** through the first cool air outlet **711**. A second cool air outlet **761** may be disposed in the second cool air duct **760**, such that cool air generated by the second evaporator **E2** is supplied to the lower storage chamber **23** through the second cool air outlet **761**.

(60) The storage chamber **20** is divided into a plurality of storage chambers by a partition **17**. The partition **17** may include a first partition **17a** to divide the storage chamber **20** into the upper storage chamber **21** and the lower storage chamber **23**, and a second partition **17b** to divide the lower storage chamber **23** into a left storage chamber **25** and a right storage chamber **26**.

(61) The upper storage chamber **21** from among two storage chambers (i.e., the upper storage chamber **21** and the lower storage chamber **23**) separated from each other by the first partition **17a** may be used as a refrigerating chamber. The lower storage chamber **23** may be divided into the left storage chamber **25** and the right storage chamber **26** by the second partition **17b**, such that the left storage chamber **25** may be used as a freezing chamber and the right storage chamber **26** may be used as both the freezing chamber and the refrigerating chamber.

(62) The above-mentioned partitioning of the storage chamber **20** is merely an example for convenience of description, and the respective storage chambers (**21**, **25**, **26**) may be used in a different way from the above-mentioned structure.

(63) A plurality of shelf units **600** may be disposed in the storage chamber **20** such that the storage chamber **20** may be divided into a plurality of storage chambers. A plurality of containers **27** to store foods therein may also be disposed in the storage chamber **20**.

(64) The opened front surface of the storage chamber **20** may be opened or closed by the door **30** rotatably coupled to the main body **10**, and a plurality of door guards **31** to store foods or the like may be mounted to the back surface of the door **30**.

(65) A hinge unit **40** may allow the door **30** to be rotatably coupled to the main body **10**. The hinge unit **40** may include an upper hinge **41** connected to an upper part of the main body **10**, an intermediate hinge **43** connected to the first partition **17a**, and a lower hinge (not shown) connected to a lower part of the main body **10**.

(66) Referring to FIGS. **1** to **3**, the insulation **15** foamed between the inner casing **11** and the outer casing **13** may be generally formed of urethane. Foaming of the insulation **15** may be performed only at a predetermined temperature or higher.

(67) Since the insulation **15** is foamed at a predetermined temperature or higher, heat is generated in the insulation foaming process, a temperature of the main body **10** may be higher than room temperature by about 20° C. or higher on the condition that the insulation **15** is foamed between the inner casing **11** and the outer casing **13**.

(68) After the insulation **15** is foamed between the inner casing **11** and the outer casing **13**, as a temperature of the main body **10** gradually drops to a room temperature, the insulation **15** is hardened or stiffened, and the main body **10** is thermally contracted.

(69) The inner casing **11** may be generally formed of a plastic material, and the outer casing **13** may be generally formed of steel. The degree of thermal contraction of the plastic material may be higher than that of the steel material by about 5 times or higher. As a result, when the main body **10** is thermally contracted, the inner casing **11** is thermally contracted much more than the outer casing **13**. Thus, during a predetermined time in which the temperature of the main body **10** drops to the room temperature, the center part of both sides of the main body **10** is modified into a convex shape in an outward direction of the main body **10**. Under the condition that the temperature of the main body **10** drops to the room temperature, the center part of both sides of the main body **10** is modified into a convex shape in an outward direction of the main body **10** such that the insulation **15** is hardened or stiffened.

(70) In order to implement the same exterior size as well as to increase inner capacity of the main body **10**, there is a need to reduce thickness of the insulation **15** foamed between the inner casing **11** and the outer casing **13**. In order to supplement insulation performance deterioration caused by thickness reduction of the insulation **15**, a vacuum insulation **19** may be disposed between the inner casing **11** and the outer casing **13**.

(71) The vacuum insulation **19** may be arranged not only in the insulation **15** foamed between the inner casing **11** and the outer casing **13** of the main body **10**, but also in the insulation **15** foamed in the door **30**. In addition, the vacuum insulation **19** may be arranged not only in the insulation **15** foamed in the partition **17**, but also in the insulation **15** foamed between the machine room cover **29** and the inner casing **11**.

(72) When deformation occurs in the inner casing **11** and the outer casing **13** due to a difference in thermal contraction between the inner casing **11** and the outer casing **13**, such deformation of the inner casing **11** and the outer casing **13** may be regularly reduced by the insulation **15** contacting each of the inner casing **11** and the outer casing **13**. If thickness of the insulation **15** is reduced, the degree of deformation in which the center part of both sides of the main body **10** is modified into a convex shape in an outward direction of the main body **10** may increase in proportion to reduced thickness of the insulation **15**.

(73) When a refrigerator starts operation upon completion of foaming of the insulation **15**, the internal temperature of the main body **10** is lowered and at the same time the degree of thermal contraction of the inner casing **11** is increased, resulting in an increase of the degree of deformation of the exterior shape.

(74) Therefore, after the insulation **15** is foamed between the inner casing **11** and the outer casing **13**, a reinforcement member **100** may be provided at both sides of the main body **10** as shown in FIGS. **4** and **5**, such that the reinforcement member **100** may prevent exterior deformation caused by a difference in thermal contraction between the inner casing **11** and the outer casing **13** when the temperature of the main body **10** drops to room temperature.

(75) The reinforcement member **100** may be formed of steel, may be disposed in the insulation **15** between the inner casing **11** and the outer casing **13** from the viewpoint of both sides of the main body **10**, and may have sufficient stiffness to prevent shape deformation caused by a difference in thermal contraction between the inner casing **11** and the outer casing **13**.

(76) The reinforcement member **100** may be arranged in the horizontal direction or the vertical direction at both sides of the main body **10** according to the flow direction of the insulation **15** foamed between the inner casing **11** and the outer casing **13**.

(77) If the insulation **15** is foamed between the inner casing **11** and the outer casing **13** and flows in the direction from the back surface to the front surface of the main body **10**, the reinforcement member **100** may be disposed in the horizontal direction at both sides of the main body **10**.

(78) If the reinforcement member **100** is disposed in the horizontal direction at both sides of the

main body **10**, the reinforcement member **100** may include a first reinforcement member **110** arranged at an upper part of the first partition **17a** on the basis of the first partition **17a** configured to divide the storage chamber **20** into an upper storage chamber **21** and a lower storage chamber **23**, and a second reinforcement member **120** disposed at a lower part of the first partition **17a**.

(79) The first reinforcement member **110** and the second reinforcement **120** may be attached to the inner casing **11** as shown in FIG. 4, and may also be attached to the outer casing **13** as shown in FIG. 6.

(80) If the first reinforcement member **110** and the second reinforcement member **120** are disposed in the insulation **15** at the spacing between the inner casing **11** and the outer casing **13**, the first reinforcement member **110** and the second reinforcement member **120** may be mounted to any of the inner casing **11** and the outer casing **13** without any problems, irrespective of the inner casing **11** and the outer casing **13**.

(81) The first reinforcement member **110** disposed at the upper part of the main body **10** may be shorter than the length of a forward-backward direction of both sides of the main body **10**, and may have thickness (T1) of about 0.5 mm.

(82) In order to increase a section modulus in a direction in which the inner casing **11** and the outer casing **13** are modified in shape at the spacing between the inner casing **11** and the outer casing **13**, it may be preferable that the first reinforcement member **110** may have the height H1.

(83) The first reinforcement member **110** may have a concavo-convex shape that is capable of having the maximum height H without disturbing flow passage of the insulation **15** foamed between the inner casing **11** and the outer casing **13**.

(84) The first reinforcement member **110** may be attached to the inner casing **11** or the outer casing **13** by an adhesive such as a double-sided tape. Although not shown in the drawings, a fixing unit to fix the first reinforcement member **110** to the inner casing **11** or the outer casing **13** may be used, such that the first reinforcement member **110** attached either to the inner casing **11** or the outer casing **13** does not move when the insulation **15** is foamed.

(85) A second reinforcement member **120** disposed at a lower part of the main body **10** may be shorter than the length of a forward-backward direction of both sides of the main body **10**, and may have thickness (T2) of about 0.5 mm.

(86) In order to increase a section modulus in the direction in which the inner casing **11** and the outer casing **13** are modified in shape at the spacing between the inner casing **11** and the outer casing **13**, the first reinforcement member **120** may have the height H2.

(87) In the same manner as in the first reinforcement member **110**, although not shown in the drawings, a fixing unit to fix the first reinforcement member **120** to the inner casing **11** or the outer casing **13** may be used, such that the first reinforcement member **110** attached either to the inner casing **11** or the outer casing **13** does not move when the insulation **15** is foamed.

(88) Referring to FIG. 7, if the insulation **15** flows from the upper part to the lower part of the main body **10** because the insulation **15** is foamed between the inner casing **11** and the outer casing **13**, a reinforcement member **130** may be vertically disposed at both sides of the main body **10**.

(89) If the reinforcement member **130** is vertically disposed at both sides of the main body **10**, the reinforcement member **130** may be shorter than the length of the vertical direction (i.e., up-and-down direction) of both sides of the main body **10**, and may have thickness of about 0.5 mm.

(90) The reinforcement member **130** vertically disposed at both sides of the main body **10** may be identical in shape to the first reinforcement member **110**, and may be longer than the length of the first reinforcement member **110** as necessary.

(91) In the same manner as in the first reinforcement member **110** and the second reinforcement member **120**, as shown in FIG. 7, the reinforcement member **130** may be attached to the inner casing **11** from among the inner casing **11** and the outer casing **13**. Although not shown in the drawings, the reinforcement member may also be attached to the outer casing from among the inner casing **11** and the outer casing **13**.

(92) As described above, as the reinforcement members (**100**, **130**) are disposed between the inner casing **11** and the outer casing **13** of both sides of the main body **10**, stiffness is increased, such that deformation of the main body **10** caused by a difference in thermal contraction between the inner casing **11** and the outer casing **13** can be reduced due to the increased stiffness.

(93) Referring to FIGS. **1** to **3**, in order to implement the same exterior size as well as to increase inner capacity of the main body **10**, thickness of the insulation **15** foamed between the inner casing **11** and the outer casing **13** must be reduced. If thickness of the insulation **15** is reduced, insulation performance may be deteriorated and stiffness is weakened, such that the main body **10** may be modified in shape by the weight of the main body **10** and the weight of materials stored in the main body **10**.

(94) In order to increase insulation performance deteriorated by the reduced thickness of the insulation **15**, the insulation **15** and a vacuum insulation panel (VIP) **19** may be disposed between the inner casing **11** and the outer casing **13**.

(95) The VIP (hereinafter referred to as a vacuum insulation) **19** may have superior insulation performance to the general insulation **15** by about 8 times or higher, and the inside of the VIP **19** is vacuumed to maximize insulation performance.

(96) Although the vacuum insulation **19** and the insulation **15** are disposed between the inner casing **11** and the outer casing **13** so as to supplement deteriorated insulation performance, it may be impossible for the vacuum insulation **19** to supplement reduced stiffness.

(97) Referring to FIGS. **8** and **9**, a reinforcement frame **200** may be provided at the front surface of the main body **10** so as to supplement reduced stiffness of the main body **10**.

(98) The reinforcement frame **200** may be disposed at the front surface of the inner casing **11** so as to supplement stiffness of the main body **10**. The reinforcement frame **200** may include an upper reinforcement frame **20** connected to an upper part of the front surface of the inner casing **11**, an intermediate reinforcement frame **220** connected to the center part coupled to a first partition **17a** mounted to the front surface of the inner casing **11**, a lower reinforcement frame **230** connected to a lower part of the front surface of the inner casing **11**, and a first side-surface reinforcement frame **240** and a second side-surface reinforcement frame **250** that are connected to both sides of the front surface of the inner casing **11**.

(99) The first side-surface reinforcement frame **240** may be disposed at the upper part of both sides of the front surface of the inner casing **11**, and an upper end of the first side-surface reinforcement frame **240** may overlap some parts of the upper reinforcement frame **210** and a lower end of the first side-surface reinforcement frame **240** may extend from the upper end to the spacing between the intermediate reinforcement frame **220** and the lower reinforcement frame **230**.

(100) The second side-surface reinforcement frame **250** may be disposed at a lower part of both sides of the front surface of the inner casing **11**, and a lower end of the second side-surface reinforcement frame **250** may be connected to the lower reinforcement frame **230** and an upper end of the second side-surface reinforcement frame **250** may extend from the lower end to a specific position spaced apart from the lower end of the first side-surface reinforcement frame **240** by a predetermined distance.

(101) Referring to FIGS. **1** to **3**, an electronic box **300** in which electronic constituent elements to control the operation of the refrigerator are contained may be provided in a forward direction of the main body **10**.

(102) Referring to FIGS. **10** to **14**, the electronic box **300** may include a base **310** to cover an electronic box installation hole **13a** provided in the forward direction of the upper part of the main body **10**, a cover **320** to cover an upper part of the base **310** in such a manner that a storage space **S** is provided at the upper part of the base **310**, a printed circuit board (PCB) **330** embedded in the storage space **S** in such a manner that electronic elements **331** are mounted thereon, a PCB mounting unit **340** to which the PCB **330** is mounted, and a reinforcement plate **350** disposed between the PCB mounting unit **340** and the cover **320**.

(103) The base **310** may include a base unit **311** connected to a front part of an upper part of the main body **10**, and a reception groove **317** to be accommodated in the electronic box installation hole **13a** when the base unit **311** is connected to the front part of the upper part of the main body **10**.

(104) The base unit **311** may form a border of the reception groove **317** having a square shape, a plurality of fixing hooks **313** may be provided not only at the border of the front part of the reception groove **317**, but also at the border of the rear part of the reception groove **317**. A wire through-hole **315** may be provided at the rear of both sides of the base unit **311**, such that a wire **333** connected to the PCB **330** is connected to the inside of the main body **10** through the wire through-hole **315**.

(105) The fixing hook **313** may include a plurality of first fixing hooks **313a** provided at the border of the front part of the reception groove **317**, and a plurality of second fixing hooks **313b** provided at the border of the rear part of the reception groove **317**.

(106) The first fixing hook **313a** may be inserted into the upper reinforcement frame **210** connected to the upper part of the front surface of the inner casing **11** and then fixed to the upper reinforcement frame **210**. The second fixing hook **313b** may be inserted into the border of the back surface of the electronic box installation hole **13a**, and may then be fixed to the border of the border of the back surface of the electronic box installation hole **13a**.

(107) Since the first fixing hook **313a** and the second fixing hook **313b** contained in the base unit **311** are respectively fixed to the upper reinforcement frame **310** and the border of the back surface of the electronic box installation hole **13a**, the base **310** connected to the front part of the upper part of the main body **10** may operate as the outer casing. When the insulation **15** is foamed between the inner casing **11** and the outer casing **13**, the base **310** does not move by the foaming pressure and may remain in a fixed state.

(108) The reception groove **317** is accommodated in the electronic box installation hole **13a** provided to the front surface of the upper part of the main body **10**, such that the reception groove **317** may be recessed at the upper part of the main body.

(109) Since the reception groove **317** is recessed at the upper part of the main body **10**, the storage space **S** disposed between the base **310** and the cover **310** may have a higher height whereas the electronic box **300** provided to the front surface of the upper part of the main body **10** may have a lower height.

(110) The cover **320** is connected to the upper part of the base **310** so that the storage space **S** may be provided between the base **310** and the cover **320**. The cover **320** may include a hinge cover unit **321** to cover the upper part of the upper hinge **41** connected to the upper part of the main body **10** in such a manner that the door **30** is rotatably coupled to the main body **10**.

(111) A plurality of PCBs **330** may be used and contained in the storage space **S** disposed between the base **310** and the cover **320**. A plurality of electronic elements **331** may be populated onto the bottom surface of each PCB **330**.

(112) The top surface of each of the PCBs **330** does not include the electronic elements **331** and is then mounted to a PCB mounting unit **340**, and the PCB mounting unit may be connected to the cover **320**.

(113) Since the PCB mounting unit **340** to which the plural PCBs **330** are mounted is connected to the cover **320**, the plural PCBs **330** may be located farthest from the upper storage chamber **21** within the storage space **S**.

(114) Since the plural PCBs **330** are located farthest from the upper storage chamber **21** within the storage space **S**, heat generated by the electronic elements **331** populated onto the plural PCBs **330** may be maximally prevented from being transferred to the inside of the upper storage chamber **21**.

(115) A connector coupling unit **341** may be provided at both sides of the PCB mounting unit **340**. A wire connector **335** to which the wire **333** connected to the PCB **330** is fixed may be connected to the connector coupling unit **341**.

(116) Therefore, the wire **333** connected to the PCB **330** may be bundled and fixed by the wire connector **335** connected to the connector coupling unit **341**, and the wire **333** bundled by the wire connector **335** may be connected to the inside of the main body **10** through the wire through-hole **315** provided to the base **310**.

(117) Accordingly, the wire **33** connected to the PCB **330** may pass through the wire through-hole **315** provided to the base **310** through both sides of the PCB mounting unit **340**. The wire **333** having passed through the wire through-hole **315** may be connected to the inside of the main body **10** through the hinge hole **41a** of the upper hinge **41**.

(118) A steel reinforcement plate **350** may be disposed between the cover **320** and the PCB mounting unit **340** to which plural PCBs **330** are mounted.

(119) When impact is applied to the upper part of the electronic box **300**, the reinforcement plate **350** reduces the impact applied to the PCBs **330** contained in the storage space **S**, resulting in protection of the electronic elements **331**.

(120) In addition, assuming that a fire breaks out in the electronic elements **331** mounted to the plurality of PCBs **330**, the reinforcement plate **350** may prevent the fire from spreading to the outside of the electronic box **330** using the reinforcement plate **350**, resulting in reduction of the possibility of causing fire accidents.

(121) Referring to FIGS. **15** to **19**, a heat dissipation pipe **400** to prevent dew formation in the outer casing **13** may be provided at the border of the front surface of the inner casing **11** of the main body **10**.

(122) If the refrigerator starts operation, cool air flows from the inside of the storage chamber **20** to the outer casing **13** forming the external appearance of the main body **10**, such that dew formation may occur in the external surface of the outer casing **13** due to a difference in temperature between the inside and the outside of the outer casing **13**.

(123) In order to prevent dew formation in the external surface of the outer casing **13**, the heat dissipation pipe **400** in which high-temperature refrigerant flows may be fixed to the border of the front surface of the inner casing **11**.

(124) A plurality of seating units **410** in which the heat dissipation pipe **400** is seated may be provided at the border of the front surface of the inner casing **11**.

(125) When the inner casing **11** is connected to the outer casing **13**, the seating unit **410** provided at the front surface border of the inner casing **11** may be located closest to the outer casing **13**.

(126) Since the seating unit **410** is located closest to the outer casing **13**, the heat dissipation pipe **400** seated in the seating unit **410** may be located farthest from the inside of the storage chamber **20**, and may be located closest to the outer casing **13**.

(127) Since the heat dissipation pipe **400** is located farthest from the inside of the storage chamber **20**, the possibility that high-temperature heat flows to the inside of the storage chamber **20** by high-temperature refrigerant flowing to the inside of the heat dissipation pipe **400** may be reduced.

(128) If high-temperature heat is transferred to the inside of the storage chamber **20**, the internal temperature of the storage chamber **20** increases by high-temperature heat, energy is unavoidably consumed by the increased temperature to reduce the internal temperature of the storage chamber **20**.

(129) The heat dissipation pipe **400** is located farthest from the inside of the storage chamber **20**, such that the possibility that high-temperature heat is transferred to the inside of the storage chamber **20** is reduced, the increasing width of the internal temperature of the storage chamber **20** is reduced, and energy consumption for reducing the internal temperature of the storage chamber **20** may be reduced.

(130) The heat dissipation pipe **400** is located closest to the outer casing **13**, and high-temperature heat caused by high-temperature refrigerant flowing in the heat dissipation pipe **400** is easily transferred to the outer casing **13**. Although cool air flows from the inside of the storage chamber **20** to the outer casing **13**, a difference in temperature between the inside and the outside is reduced,

such that dew condensation is prevented from occurring at the external surface of the outer casing **13**.

(131) The heat dissipation pipe **400** seated in the seating unit **410** may be fixed to the seating unit **410** by a plurality of clips **430**. A fixing groove **420** to which the plurality of clips is fixed may be provided in some parts of the plurality of seating units **410**.

(132) The fixing groove **410** may include a first fixing groove **421** and a second fixing groove **423** in which both ends of each clip **430** are respectively inserted. The clip **430** may include a first fixing unit **431** inserted into the first fixing groove and fixed thereto, and a second fixing unit **433** inserted into the second fixing groove and fixed thereto.

(133) The clip **430** may be fixed to the fixing groove **420** in such a manner that the heat dissipation pipe **400** is contained in the clip **430** and the heat dissipation pipe **400** is fixed to the seating unit **410**.

(134) Since the heat dissipation pipe **400** is fixed by the clip **430** on the condition that the heat dissipation pipe **400** is seated in the seating unit **410**, the heat dissipation pipe **400** may be easily fixed to the border of the front surface of the inner casing **11**.

(135) Referring to FIGS. **1** and **2**, a storage unit **500** configured to slide in the forward-backward direction may be provided in the storage chamber **20**.

(136) The storage unit **500** may be provided in the left storage chamber **25** or the right storage chamber **27** of the lower storage chamber **23**. For convenience of description, it is assumed that the storage unit **500** is provided in the right storage chamber **26**.

(137) Referring to FIGS. **20** to **27** and FIG. **31**, the storage unit **500** may include a first storage box **510** that is supported by both sidewalls of the right storage chamber **26** and slides in the forward-backward direction, a second storage chamber **520** that slides in the forward-backward direction in the inside of the first storage box **510**, and a sliding shelf **530** by which the first storage box **510** slides into and out of the right storage chamber **26**.

(138) The sliding shelf **530** may be connected to the lower part of the first storage chamber **510**, such that the first storage box **510** may slide into or out of the right storage chamber **26**.

(139) A coupling unit **26a** for coupling the rail cover **550** may be provided at both sidewalls of the right storage chamber **26**, and the coupling unit **26a** may be integrated with both sidewalls of the right storage chamber **26**.

(140) The rail cover **550** slides in the coupling unit **26a** such that the rail cover **550** is inserted into the coupling unit **26a**.

(141) The installation process of the sliding shelf **530** is as follows. The rail cover **550** of the sliding shelf **530** slides in the coupling unit **26a** and is inserted into the coupling unit **26a**, and the fastening member **B** is inserted into the fastening hole **551** provided in the rail cover **550** such that the rail cover **550** is connected to the coupling unit **26a**.

(142) If the rail cover **550** is connected to the coupling unit **26a**, the slid unit **540** slides out of the right storage chamber **26**, and the first storage box **510** is connected to the slide unit **540** in such a manner that the coupling protrusion **541a** provided to the slide unit **540** is inserted into the coupling groove **511** of the first storage box **510**.

(143) If the first storage box **510** is connected to the slide unit **540**, the slide unit **540** is guided along the rail cover **500** according to the sliding scheme, such that the first storage box **510** may slide into and out of the right storage chamber **26**.

(144) Since the sliding shelf **530** is connected to the lower part of the first storage box **510**, the first storage box **510** is completely pulled out of the right storage chamber **26**, such that a user may easily pull out foods or the like from the first storage box **510**.

(145) Since the sliding shelf **530** is connected to the lower part of the first storage box **510**, the first storage box **510** is not connected to the upper part of the sliding shelf **530**, and foods or the like may be directly stored on the sliding shelf **530**. In addition, the first storage box **510** is connected to the upper part of the sliding shelf **530**, such that foods or the like may be stored in the first storage box **510**.

(146) A detailed structure of the sliding shelf will hereinafter be given.

(147) Referring to FIGS. **20** to **27**, the sliding shelf **530** may include a rail cover **550** connected to both sidewalls of the right storage chamber **26**, a slide unit **540** configured to slide along the rail cover **550**, and a self-closing device **560** connected to the slide unit **540** so as to transfer elastic force in the insertion direction of the first storage box **510** such that the first storage box **510** can be easily closed even by weak force.

(148) The slide unit **540** may include a sliding unit **541** connected to a lower part of the first storage box **510**, and a slide rail **543** configured to slide along the rail cover **550** at both sides of the sliding unit **541**.

(149) A coupling protrusion **541a** protruding upward to interconnect the first storage box **510** and the sliding unit **541** may be provided at the upper parts of both sides of the front surface of the sliding unit **541**. A coupling groove **511** in which the coupling protrusion **541a** is inserted may be provided at the position corresponding to the coupling protrusion **541a** in the first storage box **510**.

(150) As described above, the rail cover **550** is coupled and fixed to the coupling unit **26a**, and may allow the first storage box **510** to slide into and out of the right storage chamber **26**.

(151) Referring to FIGS. **26** to **30**, the self-closing device **560** may include a case **570** respectively provided at both sides of the sliding unit **541** so as to form the external appearance, an elastic unit **580** disposed in the case **570** to accumulate elastic force when the first storage box **510** slides out as well as to transmit elastic force in the slide-in direction (i.e., the insertion direction) of the first storage box **510** when the first storage box **510** slides in, and an oil damper **590** connected to the elastic unit **580** so as to absorb impact generated when the first storage box **510** slides in.

(152) The elastic unit **580** may include a slider **581** to perform rectilinear motion within the case **570**, a rotator **683** rotatably coupled to the slider **581**, and an elastic member **585**, both ends of which are respectively connected to the slider **581** and the case **570**.

(153) The slider **581** may include a rotation hole **581a** rotatably connected to a rotation shaft **583b** mounted to the rotator **583**, a first fixing groove **581b** to which the elastic member **585** is fixed, and a second fixing groove **581c** to which the oil damper **590** is fixed.

(154) The slider **581** may perform rectilinear motion along with the rotator **583** along the following guide rail **571**. The elastic member **585** fixed to the first fixing groove **581b** of the slider **581** is extended through the rectilinear motion, such that the elastic member **585** may accumulate elastic force therein.

(155) The rotator **583** may include a protrusion unit **583a**, a rotation shaft **583b**, and an engaging groove **583c**. The protrusion unit **583a** may protrude downward from a lower part of the rotator **583** in a manner that the rotator **583** is guided along the guide rail **571** and the protrusion unit **583** is accommodated in the guide rail **571**. The rotation shaft **583b** may allow the rotator **583** to be rotatably connected to the slider **581**. The engaging groove **583c** may receive the engaging member **553** provided to the rail cover **550** such that the engaging member **553** is caught in the engaging groove **583c**.

(156) The protrusion unit **583a** may protrude toward the guide rail **571** at a lower part of the rotator **583**, and may move along the guide rail **571**, such that the rotator **583** may be guided along the guide rail **571**.

(157) The rotation shaft **583b** may be provided at the upper part of the rotator **583** so that the rotation shaft **583b** may be rotatably coupled to the rotation hole **581a** of the slide **581**.

(158) The rotator **583** may rotate about the rotation shaft **583b**, may perform rectilinear motion within a predetermined section along with the slider **581**, and may then rotate along with the slider **581**.

(159) The engaging groove **583c** in which the engaging member **553** provided to the rail cover **550** is caught is configured. When the first storage box **510** slides in or out, the rotator **583** configured to move along with the first storage box **510** may move along the guide rail **571**.

(160) The engaging member **553** provided to the rail cover **550** fixed to the coupling unit **26a** of the

right storage chamber **26** may remain in a fixed state. As a result, when the first storage box slides in or out, if the engaging member **553** is caught in the engaging groove **583c** of the rotator **583**, the rotator **583** moves along the guide rail **571**.

(161) The elastic member **585** may be implemented as a spring, and both ends of the elastic member **585** may be respectively fixed to the case **570** and the slider **581**.

(162) From among both ends of the elastic member **585**, one end fixed to the case **570** may remain in a fixed state, and the other end fixed to the slider **581** may move along with the slider **581** during the rectilinear motion of the slider **581**, may extend in length, may return to an original state, and may transmit elastic force to the first storage box **510**.

(163) The case **580** is provided at a lower part of the sliding unit **541** so as to form the external appearance, and the elastic unit **580** and the oil damper **590** may be contained in the case **580**.

(164) The case **570** may include a guide rail **571** in which the protrusion unit **583a** of the rotator **583** is accommodated and moves, a guide unit **573** acting as a passage along which the engaging member **553** moves along with the rotator **583**, a fixing unit **575** to which the elastic member **585** is fixed, a first accommodation unit **577** in which the elastic member **585** is accommodated, and a second accommodation unit **579** in which the oil damper **590** is accommodated.

(165) The guide rail **571** is designed to accommodate the protrusion unit **583a** mounted to the rotator **583** such that the protrusion unit **583a** moves along the guide rail **571**. As a result, the guide rail **571** may guide the rotator **583** and the slider **581**.

(166) The guide rail **571** may include a straight route **571a** along which the rotator **573** performs rectilinear motion in a forward-backward direction, and an engaging unit **571b** provided to one end of the straight route **571a** in a manner that the rotator **583** is rotated and fixed to the engaging unit **571b**.

(167) The guide unit **573** is arranged in parallel to the straight route **571a** of the guide rail **571a**, and is caught in the engaging groove **583c** of the rotator **583**, such that the guide unit **573** may guide the engaging member **553** configured to move along with the rotator **583** such that the engaging member **553** can perform rectilinear motion.

(168) The oil damper **590** may include a body unit **591** filled with oil and accommodated in the second accommodation unit **579** of the case **570**, and a flowing unit **593** accommodated in the body unit **591** and fixed to the second fixing groove **581c** of the slider **581** through one end thereof.

(169) Since one end of the flowing unit **593** is fixed to the slider **581**, the flowing unit **593** may move along with the slider **581**.

(170) When the first storage box **510** slides in and out, the slider **581** moves along with the first storage box **510** in the same direction as in the first storage box **510**. Therefore, when the first storage box slides in, the flowing unit **593** may slide in the body unit **591**. When the first storage box **510** slides out, the flowing unit **593** may slide out of the body unit **591**.

(171) When the flowing unit **593** slides out of the body unit **591** and then slides into the body unit **591**, impact is absorbed by oil filling the body unit **591**, such that the elastic unit **580** may be prevented from abruptly moving when the first storage box **510** slides in by elastic force of the elastic unit **580**.

(172) Therefore, impact generated when the first storage box **510** abruptly slides in by elastic force of the elastic unit **580** is absorbed, resulting in reduction of noise.

(173) The body unit **591** may maintain the condition in which the body unit **591** is accommodated in the second accommodation unit **579** of the case **570**, only the flowing unit **593** moves along with the slider **581**, and the engaging projection **579a** may be provided in the second accommodation unit **579** in such a manner that the flowing unit **593** slides into the body unit **591** and slides out of the body unit **591**.

(174) The engaging projection **579a** may be configured to have a space through which the body unit **591** does not pass and only the flowing unit **593** can pass. When the flowing unit **593** moves along with the slider **581**, the body unit **591** is caught in the engaging projection **579a** such that the

body unit **591** is prevented from moving.

(175) The first storage box **510** may slide into and out of the right storage chamber **26** by the sliding shelf **530** according to the sliding scheme.

(176) Referring to FIGS. **31** to **34**, the first storage box **510** may include a coupling groove **511** in which the coupling protrusion **531a** of the sliding shelf **530** is inserted and coupled, a guide rail **513** to guide the second storage box **520** in a manner that the second storage box **520** slides in the forward-backward direction, and a first storage box knob **515** for allowing the user to grasp the first storage box **510** such that the first storage box **510** slides into or out of the right storage chamber **26**.

(177) The guide rail **513** is provided at both sides of the inside of the first storage box **510**, such that the second storage box **520** may slide in the forward-backward direction through the guide rail **513**.

(178) The guide rail **513** may be recessed outward from the first storage box **510** from the viewpoint of both sides of the inside of the first storage box **510**.

(179) The second storage box **520** may be accommodated in the first storage box **510** and may slide in the forward-backward direction. The second storage box **520** may include a roller **521** and a second storage box knob **523**. The roller **521** may be guided along the guide rail **513** of the first storage box **510**, such that the second storage box **520** may slide in the forward-backward direction in the first storage box **510**. The second storage box knob **523** may allow the second storage box to be grasped by the user such that the second storage box **520** may move in the forward-backward direction in the first storage box **510**.

(180) The roller **521** may be provided below both sides of the outer surface of the second storage box **520**, and may be guided along the guide rail **513** provided in the first storage box **510**. A separation prevention projection **513a** may be provided at the upper part of the guide rail **513** so as to prevent separation of the roller **521**.

(181) Since the second storage box **520** is accommodated in the first storage box **510** and slides in the forward-backward direction, the guide rail **513** provided at both sides of the inside of the first storage box **510** may be spaced downward from the upper border of the first storage box **510** by a separation distance between the upper border of the second storage box **520** and the roller **521**.

(182) When the first storage box **510** slides into and out of the right storage chamber **26**, the second storage box **520** may slide into and out of the right storage chamber **26** along with the first storage box **510**, and the second storage box **520** slides in the forward-backward direction in the indoor space of the first storage box **510**, such that the inner space of the first storage box **510** may be efficiently used.

(183) Referring to FIGS. **1** and **2**, a plurality of shelf units **600** may be contained in the upper storage chamber **21**, such that the upper storage chamber **21** may be divided into a plurality of storage chambers.

(184) Referring to FIGS. **35** to **39**, the shelf unit **600** may include a shelf **610** composed of a first shelf **611** and a second shelf **613**, a bracket **620** connected to both sides of the first shelf and both sides of the second shelf **613** so as to support the first shelf **611** and the second shelf **613**, and a horizontal maintenance unit **630** provided to the bracket **620** in a manner that the first shelf **611** and the second shelf **613** are kept at a horizontal state.

(185) The shelf **610** may include a first shelf **611** located at the left of the upper storage chamber **21**, and a second shelf **613** located at the right of the upper storage chamber **21**. The first shelf **611** and the second shelf **613** may be kept horizontal, and may divide the upper storage chamber **21** into a plurality of storage chambers.

(186) A first protrusion unit **611a** may be provided at the front end of the right side of the first shelf **611**, and a second protrusion unit **613a** may be provided at the front end of the left side of the second shelf **613** so that the second protrusion unit **613a** may be spaced apart from the first protrusion unit **611a**.

(187) Under the condition that the first protrusion unit **611a** is spaced apart from the second protrusion unit **613a**, when the first shelf **611** turns right or the second shelf **613** turns left, the first protrusion unit **611a** may contact the second protrusion unit **613a**.

(188) If the first shelf **611** turns right, the first protrusion unit **611a** may contact the second protrusion unit **613a**, such that the first shelf **611** does not turn right any more. If the second shelf **613** turns left, the second protrusion unit **613a** may contact the first protrusion unit **611a**, such that the second shelf **613** does not turn left any more. As a result, the first shelf **611** and the second shelf **613** may be prevented from escaping from the horizontal state.

(189) The bracket **620** may include a first bracket **621**, a second bracket **623**, a third bracket **625**, and a fourth bracket (not shown). The first bracket **620** may be connected to the left side of the first shelf **611** so as to support the first shelf **611**. The second bracket **623** may be connected to the right side of the first shelf **611** so as to support the first shelf **611**. The third bracket **625** may be connected to the left side of the second shelf **613** so as to support the second shelf **613**. The fourth bracket (not shown) may be connected to the right side of the second shelf **613** so as to support the second shelf **613**.

(190) The bracket **620** may be supported by the support unit **640** disposed between the first cool air duct **700** and the inner casing **11** through the shelf unit fixing hole **713** provided in the first cool air duct **700**.

(191) Foods and the like may be stacked on the first shelf **611** and the second shelf **613**, and stored in the first shelf **611** and the second shelf **613**. Categories of foods stacked and stored on the first shelf **611** and the second shelf **613** may be different from each other.

(192) If the type of food stored on the first shelf **611** is different from the type of food stored on the second shelf **613**, a weight of food stored on the first shelf **611** is also different from a weight of food stored on the second shelf **613**. Accordingly, if the first shelf **611** and the second shelf **613** are used for a long period of time, the first shelf **611** and the second shelf **613** may not maintain the horizontal state and one shelf **610** from among the first shelf **611** and the second shelf **613** may be tilted downward.

(193) As described above, the horizontal maintenance unit **630** may be provided to the bracket **620** configured to support the shelf **610** such that one shelf **610** is not tilted downward and a horizontal state between the first shelf **611** and the second shelf **613** is maintained.

(194) The horizontal maintenance unit **630** may include a first fixing unit **631** and a second fixing unit **633**. The first fixing unit **631** may be connected to the second bracket **623** supporting the right side of the first shelf **611**. The second fixing unit **633** may be connected to the third bracket **625** supporting the left side of the second shelf **613**.

(195) The first fixing unit **633** and the second fixing unit **633** may be respectively connected to the second bracket **623** and the third bracket **625** by the fastening member B. The fixing protrusion **631a** may be provided in the first fixing unit **631**, and the fixing groove **633a** may be provided in the second fixing unit **633**.

(196) The first fixing unit **631** may be provided at the right side of the second bracket **623**. The second fixing unit **633** may be provided at the left side of the third bracket **625**. When a horizontal state between the first shelf **611** and the second shelf **613** is maintained, the fixing protrusion **631a** and the fixing groove **633a** may be located at matching positions.

(197) Since the fixing protrusion **631a** and the fixing groove **633a** are located at the matching positions and the fixing protrusion **631a** is inserted into the fixing groove **633a** and then fixed, the first shelf **611** and the second shelf **613** are kept in the horizontal state when the fixing protrusion **631a** is inserted into and fixed to the fixing groove **633a**.

(198) Since the fixing protrusion **631a** is inserted into and fixed to the fixing groove **633a**, only one of the first shelf **611** and the second shelf **613** is prevented from being tilted downward even when different kinds of foods are stored on the first shelf **611** and the second shelf **613** and then used for a long period of time, such that the first shelf **611** and the second shelf **613** may be kept at the

horizontal state.

(199) Referring to FIGS. 2 and 4 and FIGS. 40 to 42, the first evaporator E1 and the first blowing fan F1 that are designed to supply cool air to the upper storage chamber 21 may be disposed between the first cool air duct 700 and the inner casing 11.

(200) The first cool air duct 700 may include a front panel 710, a cool air flow passage unit 720, and a first blowing fan mounting unit 730. The front panel 710 may include a plurality of first cool air outlets 711. The cool air flow passage unit 720 may be disposed at the back surface of the front panel 710, and may form a first flow passage 725 through which cool air moves. The first blowing fan mounting unit 730 may be located at a lower part of the cool air flow passage unit 720.

(201) The front panel 710 may be formed of aluminum (Al) in a manner that the front panel 710 is evenly cooled by thermal conduction upon receiving cool air from the upper storage chamber 21, such that the indoor space of the upper storage chamber 21 may uniformly maintain a constant temperature.

(202) The front panel 710 may include a plurality of first cool air outlets 711 and a shelf unit fixing hole 713. Through the first cool air outlets 711, cool air guided through the first flow passage 725 is discharged to the inside of the upper storage chamber 21. The shelf unit fixing hole 713 may be used to fix the shelf unit 600 to the front panel 710.

(203) A lower part of the front panel 710 is formed in a streamlined shape such that the lower part of the front panel 710 is streamlined in the direction toward the upper storage chamber 21. The reason why the lower part of the front panel 710 is formed in the streamlined shape is to form the space in which the first blowing fan (F1) is installed, at the upper part of the first evaporator (E1) adjacent to the first evaporator (E1).

(204) The first blowing fan (F1) is disposed at the lower part of the front panel 710, such that the remaining parts other than the lower part of the front panel 710 may be formed in a flat panel shape.

(205) A barrier 740 may be provided at the lower part of the front panel 710, such that the barrier 740 may form the space in which the first evaporator (E1) and the first blowing fan (F1) are installed, at the lower part of the back surface of the upper storage chamber 21.

(206) Since the barrier 720 may form the space in which the first evaporator (E1) and the first blowing fan (F1) are installed, the barrier 720 may be arranged in a manner that a separation distance between the barrier 720 and the inner casing 11 is longer than a separation distance between the first cool air duct 700 and the inner casing 11.

(207) Accordingly, the upper part of the barrier 740 closely adheres to the lower part of the front panel 710 that is curved in a streamlined shape, such that the space among the first cool air duct 700, the barrier 740, and the inner casing 11 may be sealed from the upper storage chamber 21.

(208) The cool air flow passage unit 720 may include a first cool air flow passage unit 721 and a second cool air flow passage unit 723. The plurality of discharge holes 721a may correspond to the plurality of first cool air outlets 711, and may be disposed at the back surface of the front panel 710. The second cool air flow passage unit 723 is connected to the back surface of the first cool air flow passage unit 721 so that a first flow passage 725 may be formed between the first cool air flow passage unit 721 and the second cool air flow passage unit 723.

(209) The first blowing fan mounting unit 730 may be located at the lower part of the cool air flow passage unit 720. The first blowing fan mounting unit 730 may include a housing 731 mounted to the first blowing fan (F1) to be rotated, and a cover member 733 to cover the opened front surface of the housing 731.

(210) A drain unit 750 configured to drain condensed water generated from the first evaporator (E1) may be provided at a lower part of the first evaporator (E1). The drain unit 750 may have a tilted surface 751 that is gradually tilted downward with decreasing distance from the right side on the basis of the center part of the drain unit 750. A drain hole 753 may be provided at the end of the tilted surface 751.

(211) A drain pipe **755** configured to drain condensed water to the outside of the main body **10** may be provided in the drain hole **753**. The drain pipe **755** may be disposed between the inner casing **11** and the outer casing **13** of the right surface of the main body **10**.

(212) The drain pipe **755** is disposed between the inner casing **11** and the outer casing **13** of the side surface of the main body **10**, instead of being disposed between the inner casing **11** and the outer casing **13** of the back surface of the main body **10**. Therefore, when the insulation **15** is foamed in the space between the inner casing **11** and the outer casing **13** of the back surface of the main body **10**, flow of the insulation **15** may be facilitated. A structure for foaming the insulation **15** in the space between the inner casing **11** and the outer casing **13** of the back surface of the main body **10** will hereinafter be given with reference to the attached drawings.

(213) Referring to FIGS. **2** and **43**, the machine room **28** located at the lower part of the rear side of the main body **10** may be covered by a machine room cover **29**.

(214) The machine room cover **29** may include a machine room upper cover **29a** to cover the front surface and the upper part of the machine room **28**, and a machine room rear cover **29b** to cover the back surface of the machine room **28**.

(215) As can be seen from the drawings, an insulation inlet **29c** may be disposed at the position in which the insulation **15** is foamed in the space between the inner casing **11** and the outer casing **13** of the main body **10**, such that the filling space of the insulation **15** will hereinafter be referred to as the space between the inner casing **11** and the outer casing **13** for convenience of description.

(216) However, the insulation inlet **29c** may be disposed at the position in which the insulation is foamed in the inside of the door **30**.

(217) The insulation **15** is foamed in the space between the inner casing **11** and the outer casing **13** by the foam head **810**, such that the space is filled with the insulation **15**.

(218) In order to foam the insulation **15** in the space between the inner casing **11** and the outer casing **13**, the insulation inlet **29c** may be provided at the machine room's upper cover **29a** contained in the machine room covers **29** to cover the machine room **28**.

(219) From among a spacing between the inner casing **11** and the outer casing **13**, the insulation inlet **29c** may be provided at the position corresponding to a space toward the back surface of the main body **10** so as to foam the insulation **15** in the space toward the back surface of the main body **10**.

(220) The insulation inlet **29c** may be provided at the center part of the machine room cover **29** such that the insulation **15** foamed through the insulation inlet **29c** may evenly fill the space between the inner casing **11** and the outer casing **13**.

(221) In order to foam the insulation **15** in the space between the inner casing **11** and the outer casing **13**, a foam head **810** may be connected to the insulation inlet **29c** provided to the machine room's upper cover **29a**, and a guide member **820** may be connected to the insulation inlet **29c** within the space between the inner casing **11** and the outer casing **13**.

(222) The foam head **810** may allow the insulation **15** to be foamed into the insulation inlet **29c**, such that the insulation **15** may fill the space between the inner casing **11** and the outer casing **13**.

(223) Although only one insulation inlet **29c** is used and only one foam head **810** is used to correspond to the one insulation inlet **29c** as shown in the drawings, the scope or spirit of the present disclosure is not limited thereto, a plurality of insulation inlets may be used and a plurality of foam heads corresponding to the plurality of insulation inlets may also be used.

(224) If the foam head **810** is connected to the insulation inlet **29c** and the insulation **15** is then foamed through the insulation inlet **29c**, the insulation **15** is foamed in the space between the inner casing **11** and the outer casing **13** through the insulation inlet **29c** such that the space is filled with the insulation **15**. In the case of using a large-capacity refrigerator and a refrigerator in which a short distance between the inner casing **11** and the outer casing **13** is achieved such that the refrigerator includes a thin insulation thickness wall, flow of the insulation **15** is disturbed by obstacles such as a wire (not shown) contained in the space between the inner casing **11** and the

outer casing **13**, such that the discharge distance of the insulation is shortened and it is impossible to evenly fill the space between the inner casing **11** and the outer casing **13** with the insulation **15**. (225) In addition, in order to evenly fill the entire space between the inner casing **11** and the outer casing **13** with the insulation **15**, the amount of insulation **15** to be foamed in the space between the inner casing **11** and the outer casing **13** must be much larger than a volume corresponding to the space between the inner casing **11** and the outer casing **13**, such that an excess insulation **15** should be implanted in the space between the inner casing **11** and the outer casing **11**.

(226) If the insulation **15** is excessively implanted into the space between the inner casing **11** and the outer casing **11**, the hardening time of the insulation **15** foamed in the space between the inner casing **11** and the outer casing **13** may be delayed. Some parts of the insulation **15** may be exposed to the outside of the space between the inner casing **11** and the outer casing **13**, such that the exterior appearance and quality of the refrigerator may be deteriorated. The insulation **15** exposed to the outside of the space between the inner casing **11** and the outer casing **13** needs to be removed, such that the user may feel inconvenienced and the working time for filling the space between the inner casing **11** and the outer casing **13** with the insulation **15** may be extended. If the foam head **810** is unskillfully managed, there may arise a void phenomenon in which crater-shaped pores may occur in the surface of the insulator **15** hardened in the space between the inner casing **11** and the outer casing **13**.

(227) In order to address the above-mentioned issues, a guide member **820** may be used, such that the insulation **15** foamed through the foam head **810** may be guided to a part extended by a predetermined section through the space between the inner casing **11** and the outer casing **13**, instead of through the insulation inlet **29c**, without being disturbed by obstacles or the like.

(228) One end of the guide member **820** may be connected to the insulation inlet **29c** in the space between the inner casing **11** and the outer casing **13**, and the other end may be extended to the inside of the space between the inner casing **11** and the outer casing **13** such that the insulation **15** foamed through the foam head **810** may be guided.

(229) Referring to FIGS. **43** and **44**, the guide member **820** may include a connector **821** connected to the insulation inlet **29c**, and a guide pipe **823** connected to the connector **821** such that the guide pipe **823** extends to the space between the inner casing **11** and the outer casing **13**.

(230) The guide pipe **823** may be a hollow-type linear pipe, such that the insulation **15** foamed in the foam head **810** may be guided by a length of the guide pipe **823** in the space between the inner casing **11** and the outer casing **13**, without being disturbed by obstacles located in the space between the inner casing **11** and the outer casing **13**.

(231) Through the guide pipe **823**, the initial discharge position of the insulation **15** foamed in the foam head **810** may extend by the length of the guide pipe **823** in the insulation inlet **29c** within the space between the inner casing **11** and the outer casing **13**.

(232) Since the initial discharge position of the insulation extends from the lower end of the back surface of the main body **10** to the center part, disturbance of the obstacle contained in the space between the inner casing **11** and the outer casing **13** is minimized and high pressure of the insulation **15** is maintained in the guide pipe **823**, such that the entire space between the inner casing **11** and the outer casing **13** may be evenly filled with the insulation **15**, minimizing the amount of the insulation **15** to be implanted.

(233) In addition, as the insulation **15** is hardened in the space between the inner casing **11** and the outer casing **13** by surface friction generated when the insulation **15** is foamed, the void phenomenon generated on the surface of the insulation **15** may be prevented from occurring, and the amount of the insulation **15** to be implanted may be minimized. As a result the insulation **15** is not exposed to the outside so that the working time can be shortened.

(234) Referring to FIG. **45**, the guide member **830** is designed in a manner that the connector **831** and the guide pipe **833** are integrated with each other, such that the guide member **830** may be connected to the insulation inlet **29c**.

(235) Excepting that the connector **831** is integrated with the guide pipe **833**, the guide pipe **833** is implemented as a hollow-type linear pipe in the same manner as in the guide member **820** shown in FIG. **4**, and as such a detailed description thereof will herein be omitted for convenience of description.

(236) Referring to FIGS. **46** and **47**, the guide pipe **825** may include a first guide pipe **827** and a second guide pipe **829**. The first guide pipe **827** is implemented as the hollow-type linear pipe, and is connected to the connector **821**. The second guide pipe **829** may be branched from the first guide pipe **827**.

(237) The second guide pipe **829** may allow the insulation **15** having passed through the first guide pipe **827** to be branched in two directions, such that the entire space between the inner casing **11** and the outer casing **13** can be efficiently filled with the insulation **15**.

(238) The guide pipe **825** may include the first guide pipe **827** and the second guide pipe **829**, and may be formed in a Y-shape having a hollow.

(239) Referring to FIG. **48**, the guide member **830** may be designed in a manner that the connector **831** and the guide pipe **835** are integrated with each other, such that the guide member **830** may be connected to the insulation inlet **29c**. The guide pipe **835** may be formed in a Y-shape having a hollow.

(240) In the same manner as in the guide pipe **825** shown in FIG. **46**, the guide pipe **835** may be formed in a hollow-type linear pipe, such that the guide pipe **835** may include a first guide pipe **837** connected to the connector **831** and a second guide pipe **839** branched from the first guide pipe **837**.

(241) As described above, when the insulation **15** is foamed in the space between the inner casing **11** and the outer casing **13**, the guide members **820** and **830** may be used not to disturb the flow of the insulation **15**. However, instead of using the guide members **820** and **830**, the drain pipe **755** for discharging the condensed water generated from either the refrigerant pipe **P** in which refrigerant flows or the first evaporator **E1** to the outside of the main body **10** may be disposed between the inner casing **11** and the outer casing **13** of the side surface of the main body **10**. As a result, when the insulation **5** is foamed in the space between the inner casing **11** and the outer casing **13** of the back surface of the main body **10**, flow of the insulation **15** is not disturbed.

(242) Although the embodiments of the present invention have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the invention as disclosed in the accompanying claims.

MODE FOR INVENTION

(243) As described above, a related description has sufficiently been discussed in the above “Best Mode” for implementation of the embodiments of the present disclosure. Various embodiments have been described in the best mode for carrying out the invention.

Claims

1. A refrigerator comprising: a main body; a [.storage.]. .Iadd.refrigerating .Iaddend.chamber provided [.in.]. .Iadd.inside .Iaddend.the main body[, the storage chamber having a front surface that is open.]. .Iadd.and having an open front side.Iaddend.; .Iadd.a freezing chamber provided in the main body; .Iaddend. an evaporator .Iadd.configured .Iaddend.to [.supply.]. .Iadd.generate .Iaddend.cool air [.to the storage chamber.].; [.and.]. a cool air duct [.to form a flow passage to allow cool air generated by the evaporator to be supplied to the storage chamber.]. .Iadd.including: .Iaddend. [.wherein the cool air duct includes:.]. a front panel [.formed of a metal material, the front panel being disposed at a back surface of the storage chamber, such that an indoor space formed inside of the storage chamber is cooled by the cool air supplied to the storage chamber and a uniform temperature is thus maintained in the indoor space of the storage chamber.].

.Iadd.provided in the refrigerating chamber, the front panel forming a portion of a rear interior surface of the refrigerating chamber and being formed of a metal material.Iaddend., the front panel including: an upper part formed in a flat panel shape, [.and.]. a lower part.Iadd., .Iaddend.formed .Iadd.in a curved shape, that is integrally coupled .Iaddend.to [.integrally extend from.]. the upper part [.in a streamlined shape from an end of the upper part of the front panel in a direction toward the front surface of the storage chamber.], and wherein the lower part [.of the front panel.]. is formed to [.gradually curve.]. .Iadd.extend .Iaddend.outward from [.the end of the upper part of the front panel in the.]. .Iadd.a top end of the lower part in a .Iaddend.direction toward the front . [.surface.]. .Iadd.side .Iaddend.of the [.storage.]. .Iadd.refrigerating .Iaddend.chamber so that a bottom end of the lower part is [.formed to be.]. closer to the front [.surface.]. .Iadd.side .Iaddend.of the [.storage.]. .Iadd.refrigerating .Iaddend.chamber than [.a.]. .Iadd.the .Iaddend.top end of the lower part, .Iadd.a plurality of cool air outlets formed on the front panel, .Iaddend. a cool air flow passage unit disposed [.at a back surface.]. .Iadd.at a rear side .Iaddend.of the front panel . [.so as to form the flow passage.].Iadd., the cool air flow passage unit including: a first part arranged at the rear side of the front panel, and a second part coupled to the first part whereby the first part is between the front panel and the second part, the first part and the second part forming an air flow passage therebetween so that the cool air generated by the evaporator flows through the air flow passage and the cool air is discharged to the refrigerating chamber through openings of the air flow passage at corresponding positions of the plurality of cool air outlets formed on the front panel.Iaddend.; a fan mounting [.member disposed at a lower part of the cool air flow passage unit so as to supply cool air generated by the evaporator to the storage chamber.]. .Iadd.portion configured to guide the cool air generated by the evaporator through the air flow passage.Iaddend.; [.wherein the cool air flow passage unit includes: a first cool air flow passage disposed at a back surface of the front panel, and a second cool air flow passage connected to a back surface of the first cool air flow passage such that the flow passage is formed between the first cool air flow passage and the second cool air flow passage.]. .Iadd.a fan mounted in the fan mounting portion, and a barrier provided at the lower part of the front panel, the barrier forming a portion of the rear interior surface of the refrigerating chamber, wherein the evaporator and the fan are disposed at a rear side of the barrier.Iaddend..

2. The refrigerator according to claim 1, wherein the front panel is configured to have a width corresponding to a width of the [.storage.]. .Iadd.refrigerating .Iaddend.chamber.

.[3. The refrigerator according to claim 2, wherein the front panel is formed of aluminum (Al), and includes a plurality of cool air outlets through which the cool air guided through the flow passage is discharged to an inside of the storage chamber..].

4. The refrigerator according to claim [.3.]. .Iadd.1.Iaddend., [.comprising a fan, wherein the fan is arranged at an upper part of the evaporator, the lower part of the front panel is formed to accommodate an installation space of the fan.]. .Iadd.wherein the fan mounting portion includes: a housing configured to be mounted to a lower side of the cool air flow passage unit, a cover member configured to cover the housing, the cover member having an opening such that the fan is exposed through the opening of the cover member.Iaddend..

5. The refrigerator according to claim [.4.]. .Iadd.1.Iaddend., wherein [.the lower part of the front panel includes: a barrier to form a space in which the evaporator and the fan are installed, wherein the barrier closely adheres to the lower part of the front panel such that the storage chamber and the space in which the evaporator and the fan are installed are sealed up.]. .Iadd.each of the openings of the air flow passage of the air flow passage are aligned with the plurality of cool air outlets of the front panel, respectively.Iaddend..

.[6. The refrigerator according to claim 1, wherein the first cool air flow passage includes: a plurality of discharge holes located at positions corresponding to positions of the plurality of cool air outlets..].

7. The refrigerator according to claim [.6.]. .Iadd.1.Iaddend., wherein[.: the fan mounting member

is located at a lower part of the cool air flow passage unit, and includes a housing to which a fan is rotatably mounted and a cover member to cover an opened front surface of the housing.]. .Iadd.the barrier is configured to protrude toward the refrigerating chamber such that a separation distance between the barrier and the open front side is shorter than a separation distance between the front panel and the open front side.Iaddend..

8. The refrigerator according to claim .[.7, wherein: a drain member configured to drain condensed water generated from the evaporator to the outside is disposed at a lower part of the evaporator.]. .Iadd.1, further comprising a shelf unit provided in the refrigerating chamber, the shelf unit disposed above the barrier.Iaddend..

9. The refrigerator according to claim .[.8.]. .Iadd.1.Iaddend., .[.wherein.]. .Iadd.further comprising.Iaddend.: .[.the drain member has a tilted surface that is gradually tilted downward with decreasing distance from a right end of the drain member on the basis of a center part of the drain member, and a drain hole through which the condensed water is drained is provided at an end of the tilted surface.]. .Iadd.a shelf unit including a shelf and a bracket to support the shelf, a support unit disposed between the cool air duct and an inner casing of the main body to support the bracket through a fixing hole, wherein the fixing hole is disposed between the plurality of cool air outlets of the front panel.Iaddend..

10. The refrigerator according to claim .[.9.]. .Iadd.1.Iaddend., .[.wherein.]. .Iadd.further comprising.Iaddend.: .[.the drain hole includes a drain pipe through which the condensed water is drained to the outside of the main body, wherein the drain pipe is disposed between an inner casing and an outer casing of a side surface of the main body.]. .Iadd.a drain portion configured to drain condensed water generated from the evaporator which is disposed at a lower part of the evaporator, wherein the drain portion has a tilted surface gradually tilted downward, and a drain hole through which condensed water is drained is provided at an end of the tilted surface.Iaddend..

11. A refrigerator comprising: a main body; a .[.storage.]. .Iadd.refrigerating .Iaddend.chamber provided in the main body.[., the storage chamber having a front surface that is open.]. .Iadd.and having an open front side.Iaddend.; .Iadd.a freezing chamber in the main body; .Iaddend. an evaporator .Iadd.configured .Iaddend.to .[.supply.]. .Iadd.generate .Iaddend.cool air .[.to the storage chamber.].; a fan .[.to guide the cool air generated by the evaporator such that the cool air is supplied to the storage chamber; and.]. .Iadd.provided at an upper side of the evaporator and configured to blow the cool air generated by the evaporator; .Iaddend. a cool air duct .[.to form a flow passage to allow the cool air generated by the evaporator to be supplied to the storage chamber,.]. .Iadd.including: .Iaddend. .[.wherein the cool air duct includes:.]. a front panel . [formed of a metal material.]. .Iadd.provided in the refrigerating chamber.Iaddend., .Iadd.the front panel forming a portion of a rear interior surface of the refrigerating chamber, being formed of a metal material and including a plurality of cool air outlets, .Iaddend. .[.the front panel being disposed at a back surface of the storage chamber, such that an indoor space formed inside of the storage chamber is cooled by the cool air supplied to the storage chamber and a uniform temperature is thus maintained in the indoor space of the storage chamber;.]. a cool air flow passage unit disposed .[.at a back surface of the front panel so as to form the flow passage;.]. .Iadd.at a rear side of the front panel, the cool air flow passage unit including: a first part disposed at the rear side of the front panel, and a second part coupled to a rear side of the first part whereby the first part is between the front panel and the second part, the first part and the second part forming an air flow passage therebetween so that the cool air generated by the evaporator flows through the air flow passage and the cool air is discharged to the refrigerating chamber through openings of the air flow passage at corresponding positions of the plurality of cool air outlets formed on the front panel; .Iaddend.and a fan mounting .[.member that is provided at a lower part of the cool air flow passage unit, is connected to the fan.]. .Iadd.portion to which the fan is mounted, provided at a lower side of the cool air flow passage unit.Iaddend., the fan mounting . [member.]. .Iadd.portion .Iaddend.being formed to .[.protrude farther.]. .Iadd.be connectable to the

lower side of the cool air flow passage unit so that while the fan mounting portion is connected to the lower side of the cool air flow passage unit, the fan mounting portion protrudes .Iaddend.forward .[.than the cool air flow passage unit.]. .Iadd.toward the front side of the refrigerating chamber .Iaddend.where a distance between the fan mounting .[.member.]. .Iadd.portion .Iaddend.and the front .[.surface.]. .Iadd.side .Iaddend.of the .[.storage.]. .Iadd.refrigerating .Iaddend.chamber is less than a distance between the .[.cooling air.]. .Iadd.cool air .Iaddend.flow passage .Iadd.unit .Iaddend.and the front .[.surface.]. .Iadd.side .Iaddend.of the . [.storage.]. .Iadd.refrigerating .Iaddend.chamber, .Iadd.and .Iaddend. .[.wherein the cool air flow passage unit includes: a first cool air flow passage disposed at a back surface of the front panel; and a second cool air flow passage connected to a back surface of the first cool air flow passage such that the flow passage is formed between the first cool air flow passage and the second cool air flow passage.]. .Iadd.a barrier provided at a lower side of the front panel, the barrier forming a portion of the rear interior surface of the refrigerating chamber, wherein the evaporator and the fan are disposed at a rear side of the barrier.Iaddend..

12. The refrigerator according to claim 11, wherein the .[.front panel is formed of aluminum (Al), and has a width corresponding to a width of the storage chamber.]. .Iadd.the first part includes an opening arranged to correspond with a cool air outlet of the front panel, such that the cool air generated by the evaporator is discharged to an inside of the refrigerating chamber via the opening and the cool air outlet.Iaddend..

13. The refrigerator according to claim 11, wherein .Iadd.the fan mounting portion includes.Iaddend.: .[.a drain member configured to drain condensed water generated from the evaporator to the outside is disposed at a lower part of the evaporator.]. .Iadd.a housing to accommodate the fan, and a cover configured to cover the housing and having an opening such that the fan is exposed through the opening of the cover.Iaddend..

14. The refrigerator according to claim .[.13.]. .Iadd.11.Iaddend., .[.wherein: the drain member has a tilted surface that is gradually tilted downward with decreasing distance from a right end of the drain member on the basis of a center part of the drain member, and a drain hole through which the condensed water is drained is provided at an end of the tilted surface.]. .Iadd.wherein a separation distance between the barrier and the front side is shorter than a separation distance between the cool air flow passage unit and the front side.Iaddend..

15. The refrigerator according to claim .[.14.]. .Iadd.11.Iaddend., .[.wherein:.]. .Iadd.further comprising a drain member disposed at a lower part of the evaporator, the drain member configured to drain condensed water generated from the evaporator to an outside, .Iaddend. .[.the drain hole includes.]. .Iadd.wherein the drain member includes: a tilted surface gradually tilted downward; a drain hole through which the condensed water is drained, provided at an end of the tilted surface; and .Iaddend. a drain pipe .[.through which.]. .Iadd.configured to drain .Iaddend.the condensed water .[.is drained.]. to the outside of the main body.[., wherein the drain pipe is disposed between an inner casing and an outer casing of a side surface of the main body.].

.Iadd.16. A refrigerator comprising: a body including an inner casing to form a refrigerating chamber and a freezing chamber and an outer casing to form an exterior appearance of the refrigerator; an evaporator configured to generate cool air; a fan provided at an upper side of the evaporator and configured to blow the cool air generated by the evaporator; a cool air duct including: a front panel formed of a metal material, to form an upper portion of a rear interior surface of the refrigerating chamber, the front panel including a plurality of cool air outlets; a cool air flow passage unit disposed between the front panel and the inner casing, the cool air flow passage unit including: a first part disposed at a rear side of the front panel, and a second part coupled to a rear side of the first part whereby the first part is between the front panel and the second part, the first part and the second part forming an air flow passage therebetween so that the cool air generated by the evaporator flows through the air flow passage and the cool air is discharged to the refrigerating chamber through openings of the air flow passage at corresponding

positions of the plurality of cool air outlets formed on the front panel, and a fan mounting portion provided at a lower side of the cool air flow passage unit, the fan mounting portion being connectable to the cool air flow passage unit so that while the fan mounting portion is connected to the lower side of the cool air flow passage unit, the fan mounting portion is formed to protrude forward toward a front side of the refrigerating chamber where a distance between the fan mounting portion and the front side of the refrigerating chamber is less than a distance between the cool air flow passage unit and the front side of the refrigerating chamber, a barrier covering the evaporator and the fan mounting portion, the barrier forming a lower portion of the rear interior surface of the refrigerating chamber, wherein the evaporator and the fan are disposed at a rear side of the barrier. .Iaddend.

.Iadd.17. The refrigerator according to claim 16, wherein a separation distance between the barrier and the inner casing is longer than a separation distance between the cool air flow passage unit and the inner casing. .Iaddend.

.Iadd.18. The refrigerator according to claim 16, wherein the fan mounting portion includes: a housing to accommodate the fan; and a cover member configured to cover a portion of the fan, the cover member having an opening such that the fan is exposed through the opening of the cover member, wherein the cover member protrudes forward than the cool air flow passage unit. .Iaddend.

.Iadd.19. The refrigerator according to claim 16, wherein the first part includes a plurality of openings arranged to correspond with a plurality of cool air outlets of the front panel, such that the cool air generated by the evaporator is discharged to an inside of the refrigerating chamber via the plurality of openings and the plurality of cool air outlets. .Iaddend.

.Iadd.20. The refrigerator according to claim 16, further comprising: a drain portion configured to drain condensed water generated from the evaporator which is disposed at a lower part of the evaporator, wherein the drain portion has a tilted surface gradually tilted downward, and a drain hole through which condensed water is drained is provided at an end of the tilted surface. .Iaddend.

.Iadd.21. The refrigerator according to claim 16, further comprising a shelf unit to divide the refrigerating chamber horizontally and a support unit disposed between the cool air duct and the inner casing of the body, wherein the shelf unit is supported by the support unit via a shelf unit fixing hole of the cool air duct. .Iaddend.

.Iadd.22. The refrigerator according to claim 21, wherein the air flow passage includes a first air flow passage and a second air flow passage that are formed such that the shelf unit fixing hole is disposed between the first air flow passage and the second air flow passage. .Iaddend.
